NNN-C-1698A
June 12, 1975
SUPERSEDING
Int. Fed. Spec.
NNN-C-001698 (DSA-DM)
July 19, 1971 and

FEDERAL SPECIFICATION

COUNTER, BLOOD CELLS, Differential

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a differential counter suitable for use in routine white blood cell counts.

1.2 Classification.

1.2.1 Sizes. Differential blood cell counters shall be of the following sizes:

Size 5 - Five units and totalizer.
Size 8 - Eight units and totalizer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.

Federal Standard

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

FSC 6640

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Military Specifications

MIL-M-7298	Manual, Technical: Commercial Equipment.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The material shall be of high commercial quality throughout and suitable in every respect for the purpose intended.

3.2 Description. Each differential blood cell counter shall be manually operated and shall be a combination adding machine and percentage calculator. Blood cell counter shall consist of a series of counting units, each independently operated by a key and joined together in a line: individual amount windows for each key; a totalizer: a removable blood cell chart; and a reset device. The counter shall be suitable for automatically computing and tabulating different blood cells. Operating mechanism of each counting unit shall be housed in a suitable casing closed at

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all points. Housing shall not interfere with the working of the counter. The relationship of the face of the number drum and the windows in the unit housing shall be such that the counter can be read when placed on a table while the operator is sitting or standing. The counter body and gears shall be constructed of metal and all parts subject to extreme wear shall be case-hardened; all external parts shall be of firm construction so as not to break off easily; and all moving parts shall be restrained to prevent tangling of the internal mechanism.

3.3 Counter body. Each counter body shall be constructed of metal. The operating mechanism of each unit shall be housed in a casing closed at all points. The housing shall not interfere with the working of the counter.

3.4 Keys. The keys shall be constructed of metal. Tops of keys may be constructed of a suitable plastic. A single complete stroke of the key shall result in accurate counting and there shall be no double counting, skipping or coasting. The counter shall remain stable on the table top when the keys are being operated. The number of times each individual key is depressed shall register independently in its amount window. The windows shall be covered with clear glass or clear plastic. The totals of the individual keys shall automatically record percentages when a total of 100 cells has been counted.

3.4.1 Size 5. For size 5, each window shall register up to 999.

3.4.2 Size 8. For size 8, windows shall register to 9, 9, 9, 9, 99, 999, 99 and 99 respectively, or each window may register to 999.

3.5 Totalizer. Each counter shall be equipped with a totalizer which will register the total number of times all of the individual keys have been depressed. The counter shall indicate that the totalizer has reached a count of 100 by the signalling of a bell.

3.6 Reset device. The reset device shall simultaneously reset all of the key counters and the totalizer to zero by one revolution.

3.7 Classification strip. Each counter shall be provided with a removable classification strip or blood cell chart made of durable material. The chart shall show the name and illustration of the particular blood cell type to be counted on each key. Each blood cell type shall be permanently and legibly illustrated in color simulating morphological characteristics of the blood cells. The chart shall be fixed to the body of the counter in a readable position for the operator.

3.7.1 Size 5. The arrangement of cell types for size 5 shall be as specified.

3.7.2 Size 8. Unless otherwise specified for size 8, the arrangement from left to right of the cell types shall be as follows:

Baso. Eosin. Myelo. Juven. Stab. Seg. Lymph. Mono.

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3.8 Sizes.

3.8.1 Size 5. Each size 5 counter shall have five individual counting units and a totalizer.

3.8.2 Size 8. Each size 8 counter shall have eight individual counting units and a totalizer.

3.9 Accuracy. The counter shall be accurate when tested as specified in 4.4.1.

3.10 Endurance. The counter shall show no breakdown of any component part and indicate an accurate count upon completion of test when tested as specified in 4.4.2.

3.11 Finish. Unless otherwise specified, manufacturer's standard finish shall be acceptable.

3.12 Service data. The contractor shall furnish with each unit two copies each of a technical manual which contains complete instructions for installation, operati and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on the maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.13 Identification marking. Each counter shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.14 Workmanship. The workmanship shall be first class in every respect. The finished blood cell counters shall be free from defects listed in Table III and other defects which detract from their appearance or which may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

Table I Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Acceptance number shall be zero.

Table II Sampling for tests

Unit of product	Characteristic	Requirement	Test procedure	Inspection level
Counter	Accuracy	3.9	4.4.1	S-2
	Endurance	3.10	4.4.2	S-2

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4.3 Examination. The blood counter shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with but shall not be limited to defects listed in Table III.

Table III Classification of defects

Categories	Defects
Major	
101	Component missing.
102	Finish not uniformly applied; coating not free of peeling or chipping.
103	Counter not free of corroded areas.
104	Counter not free of sharp edges or burrs.
105	Classification strip not free of tears or defacing.
106	Cell illustration in classification strip incorrect, indistinguishable, or not colored.
107	External parts not free of damage.
108	Windows not free of cracks or tears.
109	Reset knobs not secure (loose).
110	Instructions not adequate, accurate, complete or present.
111	Name of cells on illustration strips missing.
112	Signal bell does not ring clearly (muffled).
113	Illustration of cells not colored.
Minor	
201	Item identification missing, illegible, or incomplete.
202	Windows not clear (hazy, marred).
203	Classification strip not secure.
204	Housing not free of indentations.

4.4 Tests.

4.4.1 Accuracy test. In testing the performance of counters in 4.4.1.1 and 4.4.1.2, keys shall be pressed so as to advance all registering wheels.

4.4.1.1. Varying number wheel key control. On counters having a variation in number of wheels controlled by each key, depress the key that controls one wheel nine times, depress the key that controls two wheels 99 times and depress the key that controls three wheels 999 times (see Table IV) observing the totalizer reading at the completion of each key actuation. Repeat four times without resetting counters, observing totalizer reading at end of each complete cycle.

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Table IV Cycles

Key	Key pressed	1	2	3	4	5
1	9	9	1341	2673	4005	5337
2	9	18	1350	2682	4014	5346
3	9	27	1359	2691	4023	5355
4	9	36	1368	2700	4032	5364
5	99	135	1467	2799	4131	5463
6	999	1134	2466	3798	5130	6462
7	99	1233	2565	3897	5229	6561
8	99	1332	2664	3996	5328	6660
Totalizer * window reading (completed cycles)		332	664	996	328	660

4.4.1.2 Three-wheel key control. On counters in which all keys control three wheels, depress each key in turn 99 times observing the totalizer reading at the completion of each key actuation. Repeat four times without resetting counters, observing totalizer reading at end of each complete cycle (see Table V).

Table V Cycles

Key	Key pressed	1	2	3	4	5
1	99	99	891	1683	2475	3267
2	99	198	990	1782	2574	3366
3	99	297	1089	1881	2673	3465
4	99	396	1188	1980	2772	3564
5	99	495	1287	2079	2871	3663
6	99	594	1386	2178	2970	3762
7	99	693	1485	2277	3069	3861
8	99	792	1584	2376	3168	3960
Totalizer * window reading (completed cycles)		792	584	376	168	960

* Totalizer window shows only last three figures of total count for each complete cycle.

4.4.2 Endurance. The counter shall be set up on an endurance cycling device designed to depress all keys simultaneously at a rate of 120 cycles per minute. The plungers shall be so adjusted that the key depression fully actuates the counter but stops short of full depression by at least 1/16 inch. The counter shall be actuated 15,000 depressions.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

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5.1.1 Unit of issue. One counter, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units indicated in Table VI shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VI Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Box shall contain interior fittings or cushioning material to insure against shifting of or damage to the item. Closure shall be adequate to prevent accidental opening. As an alternate, a light wood box may be used in lieu of the fiberboard box.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard

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boxes. Fiberboard box shall be waterproof in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Size required (see 1.2.1).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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6.3 This specification is applicable to the following item listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6630-00-418-8010	COUNTER, BLOOD CELLS, Differential

Custodians:
Army - MD
Navy - MS
Air Force - 03

Civil agencies interested:
GSA-FSS

Preparing activity:
Defense Supply Agency - DM

Project No. 6630-0228

U. S. GOVERNMENT PRINTING OFFICE : 1975 - 210-814/901

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DOCUMENT IDENTIFIER AND TITLE
NNN-C-1698A - Counter, Blood Cells, Differential

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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